Data collection

Experimental

i.e. Dealing with collected data.

Chp1: Data

Methods of sampling, types of data, and populations vs samples.

Chp2: Measures of Location/Spread

Statistics used to summarise data, including mean, standard deviation, quartiles, percentiles. Use of linear interpolation for estimating medians/quartiles.

Chp3: Representation of Data

Producing and interpreting visual representations of data, including box plots and histograms.

Chp4: Correlation

Measuring how related two variables are, and using linear regression to predict values.

Theoretical

Deal with probabilities and modelling to make inferences about what we 'expect' to see or make predictions, often using this to reason about/contrast with experimentally collected data.

Chp5: Probability

Venn Diagrams, mutually exclusive + independent events, tree diagrams.

Chp6: Statistical Distributions

Common distributions used to easily find probabilities under certain modelling conditions, e.g. binomial distribution.

Chp7: Hypothesis Testing

Determining how likely observed data would have happened 'by chance', and making subsequent deductions.

Interestingly, most of this chapter is from the old S3 module (a Further Maths module!) with also some S2. There is little 'calculation' involved in this chapter; consider this a 'bookwork' one!

1: Populations vs samples

"Suggest why we would not test all the light bulbs."
"Identify the sampling frame."

2: Random Sampling

Describe the disadvantages of systematic sampling.

3: Non-Random Sampling

Describe how a stratified sample would be conducted, including strata sizes.

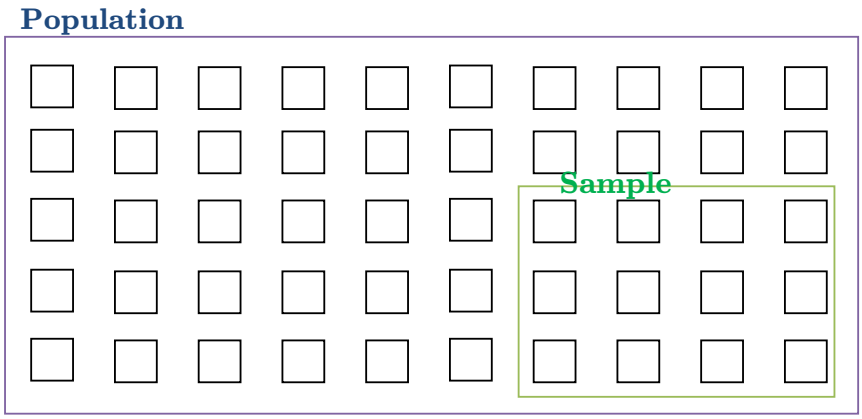
4: Types of data

Continuous vs discrete, terms such as class intervals, class boundaries, class width.

5: Edexcel's 'Large Data Set'



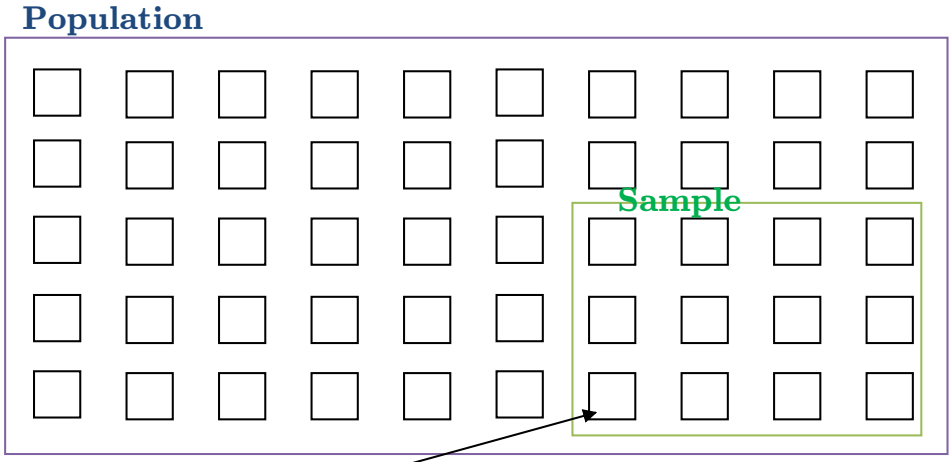
What you're expected to know about the 'large data set' of weather data, and how to use it.



A population is:	?
A sample is:	?

You're probably used to a 'population' meaning all humans/animals within a country/ecosystem. But a population could be "*all the lightbulbs in a factory*" or "*all the cars in the UK*".

Sampling key terms 5



Each individual thing in the population that can be sampled is known as a **sampling unit**.

Often sampling units of a population are individually named or numbered **to form a list** called the **sampling frame**.

Populations vs Samples 6

We could collect data either from a sample, or from the entire population.
Data collected from the entire population is known as a **c**?

	Advantages	Disadvantages
Census	?	?
Sample	?	?

Example: A supermarket wants to test a delivery of avocados for ripeness by cutting them in half.

a. Suggest a reason why the supermarket should not test all the avocados in the delivery.

b. The supermarket tests a sample of 5 avocados and finds that 4 of them are ripe. They estimate that 80% of the avocados in the deliver are ripe. Suggest one way that the supermarket could improve their estimate.

a ?

b ?

1 A school uses a census to investigate the dietary requirements of its students.

a Explain what is meant by a census.

b Give one advantage and one disadvantage to the school of using a census.

a?

b?

2 A factory makes safety harnesses for climbers and has an order to supply 3000 harnesses. The buyer wishes to know that the load at which the harness breaks exceeds a certain figure.

a Suggest a reason why a census would not be used for this purpose.

The factory tests four harnesses and the load for breaking is recorded:

320 kg 260 kg 240 kg 180 kg

b The factory claims that the harnesses are safe for loads up to 250 kg. Use the sample data to comment on this claim.

c Suggest one way in which the company can improve their prediction.

a?

b?

c?

3 A city council wants to know what people think about its recycling centre.
The council decides to carry out a sample survey to learn the opinion of residents.

a Write down one reason why the council should not take a census.

b Suggest a suitable sampling frame.

c Identify the sampling units.

a?

b?

c?

5 A manager of a garage wants to know what their mechanics think about a new pension scheme designed for them. The manager decides to ask all the mechanics in the garage.

a Describe the population the manager will use.

b Write down the main advantage in asking all of their mechanics.

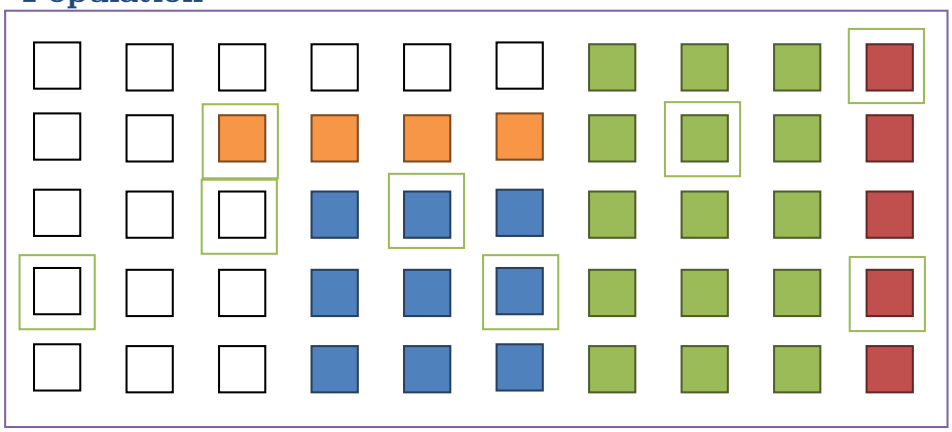
a?

b?

Lay out your notes like this for next bit of the chapter. Use a full page.

	Type	How to carry out	Advantages	Disadvantages
Random Sampling	Simple Random Sampling			
	Systematic Sampling			
	Stratified Sampling			
Non-Random	Quota Sampling			
	Opportunity Sampling			

Population



Ordinarily, we would want each thing in our sampling frame to have an **equal chance of being chosen**, in order to **avoid bias**.

This is known as **random sampling**.
There are a few ways of doing this. . .

Simple Random Sampling 12

Type	How to carry out	Advantages	Disadvantages
Simple Random Sampling	**What is it :** ? **Method:** ?	?	?

Edexcel S3 June 2004 Q1a

There are 64 girls and 56 boys in a school. Explain briefly how you could take a random sample of 15 pupils using a simple random sample. **(3)**

?

Type	How to carry out	Advantages	Disadvantages
Systematic Sampling	**What is it :** ?	?	?

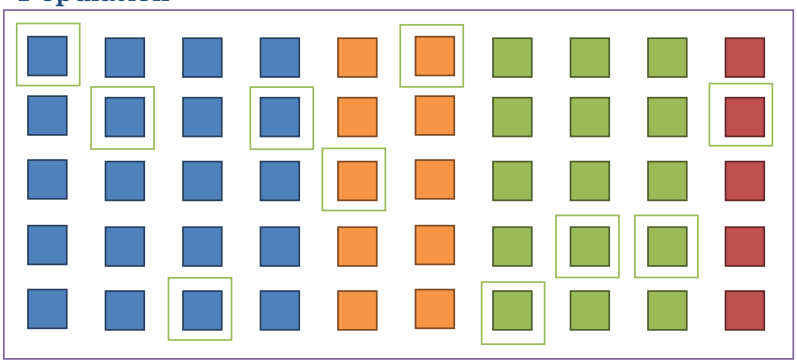
Edexcel S3 June 2009 Q1a

A telephone directory contains 50 000 names. A researcher wishes to select a systematic sample of 100 names from the directory. Explain in detail how the researcher should obtain such a sample. **(2)**

?

We want to sample 20% of the population. If the population were divided into distinct groups (e.g. age ranges), known as '**strata**', we could randomly sample 20% from each group, ensuring each group is equally represented.

Population



Type	How to carry out	Advantages	Disadvantages
Stratified Sampling	**What is it :** ?	?	?

Edexcel S3 Jan 2006 Q1

A school has 15 classes and a sixth form. In each class there are 30 students. In the sixth form there are 150 students. There are equal numbers of boys and girls in each class. There are equal numbers of boys and girls in the sixth form. The head teacher wishes to obtain the opinions of the students about school uniforms. Explain how the head teacher would take a stratified sample of size 40. **(7)**

?

You would certainly want to know your mark scheme on this one!

2 A survey is carried out on 100 members of the adult population of a city suburb. The population of the suburb is 2000. An alphabetical list of the inhabitants of the suburb is available.

a Explain one limitation of using a systematic sample in this situation.

b Describe a sampling method that would be free of bias for this survey.

a?

b?

3 A gym wants to take a sample of its members. Each member has a 5-digit membership number, and the gym selects every member with a membership number ending 000.

a Is this a systematic sample? Give a reason for your answer.

b Suggest one way of improving the reliability of this sample.

a?

b?

4 A head of sixth form wants to get the opinion of year 12 and year 13 students about the facilities available in the common room. The table shows the numbers of students in each year.

	Year 12	Year 13
Male	70	50
Female	85	75

a Suggest a suitable sampling method that might be used to take a sample of 40 students.

b How many students from each gender in each of the two years should the head of sixth form ask?

a?

b?

5 A factory manager wants to get information about the ways their workers travel to work. There are 480 workers in the factory, and each has a clocking-in number. The numbers go from 1 to 480. Explain how the manager could take a systematic sample of size 30 from these workers.

?

6 The director of a sports club wants to take a sample of members. The members each have a unique membership number. There are 121 members who play cricket, 145 members who play hockey and 104 members who play squash. No members play more than one sport.

a Explain how the director could take a simple random sample of 30 members and state one disadvantage of this sampling method.

The director decides to take a stratified sample of 30 members.

b State one advantage of this method of sampling.

c Work out the number of members who play each sport that the director should select for the sample.

a?

b?

c?

Consider the following scenario: You wish to conduct a survey in the UK **on whether being left-handed affects IQ**. We need to choose people to assess.

Why would random sampling be problematic? **Because we don't know the sampling**

?

Famous Lefties



For this scenario we'd likely use **quota sampling**, i.e.

1. As with stratified sampling, divide population into groups according to characteristic of interest, then determine size of each group in sample to reflect proportions within the population.
2. But instead of random sampling within each group, we actively choose people within each group via suitable means (e.g. advertising), **until the 'quota' for each group is filled**.

A variant of this is **opportunity sampling**, where we find people **at the same time the survey is being carried out** (e.g. exit polls at polling stations). This is not a suitable method for the left-handed example, because giving the likely time-consuming nature of assessment coupled with resources required, we'd likely arrange with the people taking part before the actual assessment tasks took place.

Type	How to carry out	Advantages	Disadvantages
Quota Sampling	?	?	?
Opportunity/ Convenience Sampling	?	?	?

Edexcel S3 June 2010 Q2

A lake contains 3 species of fish. There are estimated to be 1400 trout, 600 bass and 450 pike in the lake. A survey of the health of the fish in the lake is carried out and a sample of 30 fish is chosen.

(a) Give a reason why stratified random sampling cannot be used. **(1)**

(b) State an appropriate sampling method for the survey. **(1)**

(c) Give one advantage and one disadvantage of this sampling method. **(2)**

(d) Explain how this sampling method could be used to select the sample of 30 fish. You must show your working. **(4)**

(a) ?

(b) ?

(c) ?

(d) ?

1 Interviewers in a shopping centre collect information on the spending habits from a total of 40 shoppers.

a Explain how they could collect the information using:
i quota sampling **ii** opportunity sampling

b Which method is likely to lead to a more representative sample?

a(i)?

a(ii)?

b?

2 Describe the similarities and differences between quota sampling and stratified random sampling.

?

3 An interviewer asks the first 50 people he sees outside a fish and chip shop on a Friday evening about their eating habits.

a What type of sampling method did he use?

b Explain why the sampling method may not be representative.

c Suggest two improvements he could make to his data collection technique.

a?

b?

c?

4 A researcher is collecting data on the radio-listening habits of people in a local town. She asks the first 5 people she sees on Monday morning entering a supermarket. The number of hours per week each person listens is given below:

4 7 6 8 2

a Use the sample data to work out a prediction for the average number of hours listened per week for the town as a whole.

b Describe the sampling method used and comment on the reliability of the data.

c Suggest two improvements to the method used.

a?

b?

c?

5 In a research study on the masses of wild deer in a particular habitat, scientists catch the first 5 male deer they find and the first 5 female deer they find.

a What type of sampling method are they using?

b Give one advantage of this method.

The masses of the sampled deer are listed below.

Male (kg)	75	80	90	85	82
Female (kg)	67	72	75	68	65

c Use the sample data to compare the masses of male and female wild deer.

d Suggest two improvements the scientists could make to the sampling method.

a?

b?

c?

d?

6 The heights, in metres, of 20 ostriches are listed below:

1.8, 1.9, 2.3, 1.7, 2.1, 2.0, 2.5, 2.7, 2.5, 2.6, 2.3, 2.2, 2.4, 2.3, 2.2, 2.5, 1.9, 2.0, 2.2, 2.5

a Take an opportunity sample of size five from the data.

b Starting from the second data value, take a systematic sample of size five from the data.

c Calculate the mean height for each sample.

d State, with reasons, which sampling method is likely to be more reliable.

a?

b?

c?

d?

Qualitative/Categorical

Non-numerical values, e.g. colour.

Quantitative

Numerical values.

Discrete

Can only take specific values, e.g. shoe size, number of children.

Note that while discrete variables only allow specific values, the range could still be infinite, e.g. "number of attempts before success".

Continuous

Can take any decimal value (possible with a specified range).

Weight **w** (kg)	Frequency
$0 \leq w < 20$	3
$20 \leq w < 70$	4

Data can be **grouped** for conciseness, at the expense of losing the exact original values.

This is known as a ?

$20 \leq w < 70$

? ? ?

Midpoint = 4? ?

Class width = ?

1 State whether each of the following variables is qualitative or quantitative.

a Height of a tree

b Colour of car

c Time waiting in a queue

d Shoe size

e Names of pupils in a class

a?

b?

c?

d?

e?

2 State whether each of the following quantitative variables is continuous or discrete.

a Shoe size

b Length of leaf

c Number of people on a bus

d Weight of sugar

e Time required to run 100 m

f Lifetime in hours of torch batteries

a?

b?

c?

d?

e?

f?

3 Explain why:

a 'Type of tree' is a qualitative variable

b 'The number of pupils in a class' is a discrete quantitative variable

c 'The weight of a collie dog' is a continuous quantitative variable.

a?

b?

c?

4 The distribution of the masses of two-month-old lambs is shown in the grouped frequency table.

a Write down the class boundaries for the third group.

b Work out the midpoint of the second group.

c Work out the class width of the first group.

Mass, m (kg)	Frequency
$1.2 \leqslant m < 1.3$	8
$1.3 \leqslant m < 1.4$	28
$1.4 \leqslant m < 1.5$	32
$1.5 \leqslant m < 1.6$	22

a?

b?

c?

Simple Random Sampling	Systematic Sampling	Stratified Sampling	Quota Sampling	Opportunity Sampling

Suggest a suitable sampling method.

"You wish to test lightbulbs produced by a factory in a daily batch."

?

"You wish to survey consumer opinion on your new drink *FizzGuzz* released in the UK."

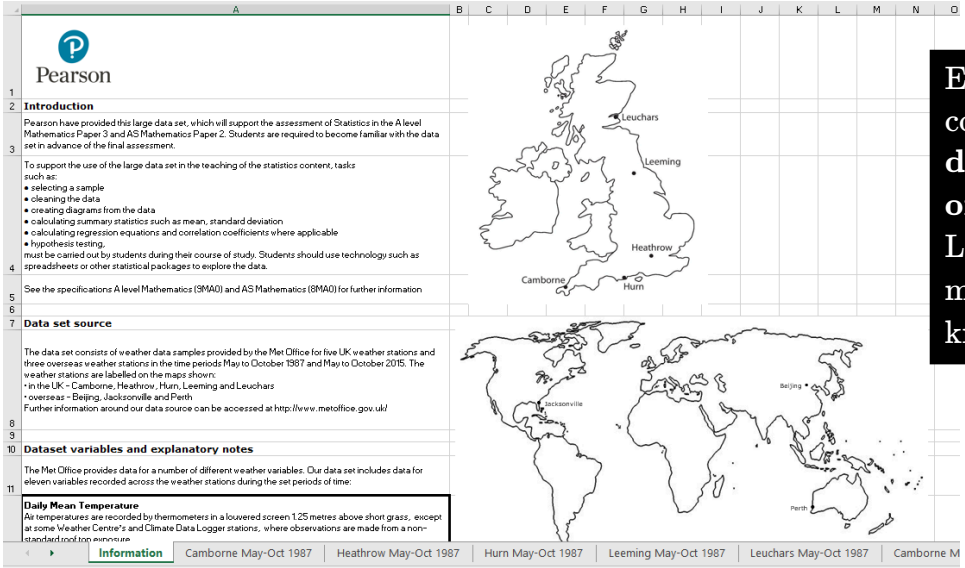
?

"You wish to determine students' favourite TV programmes in your school, that is fairly representative of each year group."

?

All A Level exam boards are obligated to provide a 'large data set'. Data in exam questions will often be from this set, and you are encouraged to explore this data (which is publicly available) in Microsoft Excel.

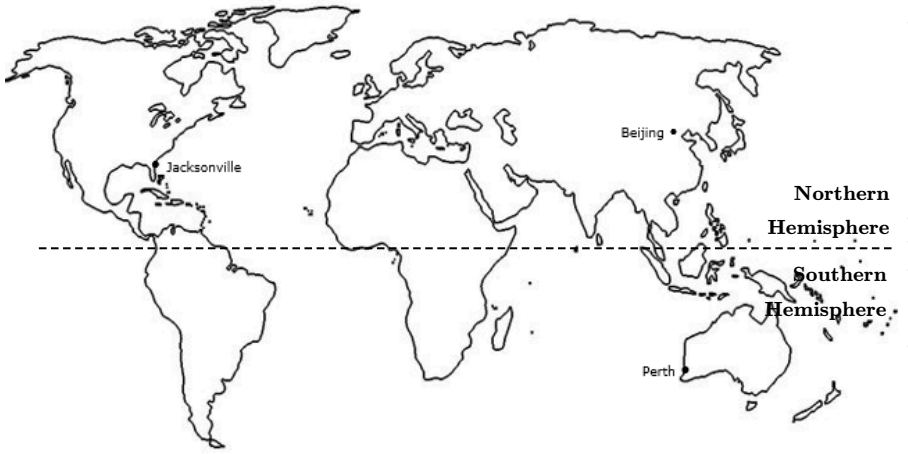
It is important to note that you are expected to be familiar with this data set before you go into your exam, including some basic geographic knowledge!



Edexcel's data set concerns **weather data from a number of weather stations**. Let's explore what you might be expected to know. . .

https://qualifications.pearson.com/content/dam/pdf/A%20Level/Mathematics/2017/specification-and-sample-assesment/Pearson%20Edexcel%20GCE%20AS%20and%20AL%20Mathematics%20data%20set%20-%20Issue%201%20(1).xls





1 You should know the names and rough locations of the 5 UK weather stations, as well as the 3 international weather stations.

The data was recorded for:

- May-Oct 1987
- May-Oct 2015

2 You should be familiar with the variables involved and their respective units.

Total rainfall

(in mm)

tr/trace means less than 0.05mm

Mean Windspeed

kn/knot is "nautical mile per hour". $1kn = 1.15mph$

Windspeed also given on **Beaufort Scale**:

0 = Calm	$< 1kn$
1-3 = Light	1-10kn
4 = Moderate	11-16kn
5 = Fresh	17-21kn

Mean Visibility

How far (in metres) can be seen into the horizon during daylight hours.

Wind Direction

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	CAMBORNE		© Crown Copyright Met Office 2015												
2	NGR = 1627E 4067N														
3	Altitude = 87 metres														
4	Latitude = 50:22N Longitude = 05:33W														
5															
6	Date	Daily Mean Temperature (0900-0900) (°C)	Daily Total Rainfall (0900-0900) (mm)	Daily Total Sunshine (0000-2400) (hrs)	Daily Mean Windspeed (0000-2400) (kn)	Daily Mean Windspeed (0000-2400) (Beaufort conversion)	Daily Maximum Gust (0000-2400) (kn)	Daily Maximum Relative Humidity %	Daily Mean Total Cloud (oktas)	Daily Mean Visibility (Dm)	Daily Mean Pressure (hPa)	Daily Mean Wind Direction (o)	Cardinal Direction	Daily Max Gust Corresponding Direction (o)	Cardinal Direction
7	01/05/1987	10.7	3.1	n/a	n/a	n/a	n/a	100	7	2000	1018	360	N	20	NNE
8	02/05/1987	8.9	0.1	n/a	n/a	n/a	n/a	91	3	3200	1020	320	NW	330	NNW
9	03/05/1987	8.1	0	n/a	n/a	n/a	n/a	77	5	3600	1029	350	N	350	N
10	04/05/1987	8.2	0	n/a	n/a	n/a	n/a	83	5	4100	1036	350	N	350	N
11	05/05/1987	9.8	0	n/a	n/a	n/a	n/a	86	5	2700	1036	10	N	10	N
12	06/05/1987	9.3	0	n/a	n/a	n/a	n/a	100	1	1000	1033	330	N	340	NNW
13	07/05/1987	10.9	0	n/a	n/a	n/a	n/a	100	3	600	1031	350			
14	08/05/1987	10.5	tr	n/a	n/a	n/a	n/a	89	1	2400	1025	110			
15	09/05/1987	10.9	0	n/a	n/a	n/a	n/a	95	3	900	1017	360			
16	10/05/1987	9.9	0	n/a	n/a	n/a	n/a	79	4	4100	1018	10			
17	11/05/1987	8.8	6	n/a	n/a	n/a	n/a	95	7	2500	1017	270	W	260	W
18	12/05/1987	10.2	tr	n/a	n/a	n/a	n/a	97	5	2400	1009	310	NW	310	NW
1			2.2	n/a	n/a	n/a	n/a	77	4	4600	1016	340	NNW	340	NNW
2			tr	5.9	16	Moderate	35	95	7	3100	1008	290	WNW	270	W
2			0	12.3	13	Moderate	27	77	4	4500	1012	10	N	10	N
2			tr	11.6	6	Light	16	92	4	3700	1015	290	WNW	290	WNW

Mean Pressure

In hectopascals (hPa)

Mean temperature

(in $^{\circ}C$)

Textbook claims this is **max** temp for UK, but it is **mean** temp for all locations.

Total sunshine

(nearest $\frac{1}{10}$ of an hour)

Maximum Gust

(in kn) is highest instantaneous wind speed.

Humidity

is the % of air saturation with water vapour. 100% is the maximum % water content air can contain.

Mean Cloud Cover

Oktas means the number of $\frac{1}{8}$ths of the sky covered.

3 You should have a vague idea of the range of values for each location.

UK Location (2015)	Temp Range	Wind Speed Range
Camborne	10-20	3-18
Heathrow	8-**29**	3-19
Hurn	6-24	2-19
Leeming	4-23	3-17
Leuchars	4-19	3-23

World Location (2015)	Temp Range	Wind Speed Range
Beijing	8-33	2-9
Jacksonville	15-31	1-12
Perth	8-25	4-14

Beijing temp range relatively large.
Min Jacksonville temp high.
Perth similar to UK.

Mean wind speed in UK across full period was roughly 9 nm. But 4 nm in Beijing (i.e. lower), 5 in Jacksonville (again lower), 8 in Perth (similar to UK).

From new A Level sample assessment materials:

"A meteorologist believes that there is a relationship between the daily mean windspeed, w kn, and the daily mean temperature, t °C. A random sample of 9 consecutive days is taken from past records from a town in the UK in July and the relevant data is given in the table below. . . .

Using the same 9 days, a location from the large data set gave $\overline{t} = \mathbf{27.2}$ and $\overline{w} = \mathbf{3.5}$.

(d) Using your knowledge of the large data set, suggest, giving your reason, the location that gave rise to these statistics."

(Note to teachers: I will not otherwise use SAM questions in these slides. I made one exception here!)

4 You should have a vague idea of the range of values for each variable for the data set as a whole.

Variable	Typical value(s)
Gust (UK only)	8 – 52 nm
Rainfall	0 – 60 mm in UK, but more extreme maximums elsewhere (e.g. 102mm in Perth)
Pressure	988 – 1038 hPa
Wind Speed on Beaufort scale	Max is 'fresh' (5). Most Light or Moderate.
Sunshine (UK only)	0 – 16 hrs
Cloud Cover	0 – 8 ocktas (i.e. full spread)

Hurn

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Date	Daily Max Temp (09-00-0900 C)	Daily Total Rainfall (0900-0900) (mm)	Daily Total Sunshine (0000-2400) (hrs)	Daily Mean Windspeed (0000-2400) (kn)	Daily Mean Windspeed (0000-2400) (Beaufort conversion)	Daily Maximum Gust (0000-2400) (kn)
01/06/1987	15.1	0.6	4.5	7	Light	19
02/06/1987	12.5	4.7	0	7	Light	22
03/06/1987	13.8	tr	5.6	11	Moderate	25
04/06/1987	15.5	5.3	7.8	7	Light	17
05/06/1987	13.1	19	0.5	10	Light	33
06/06/1987	13.8	0	8.9	19	Fresh	46
07/06/1987	13.2	tr	3.8	11	Moderate	27
08/06/1987	12.9	1	1.7	9	Light	19
09/06/1987	11.2	tr	5.4	6	Light	19
10/06/1987	9.2	1.3	9.7	4	Light	n/a
11/06/1987	12.6	0	12.5	6	Light	18
12/06/1987	10.4	0	11.9	5	Light	n/a
13/06/1987	9.6	0	8.6	5	Light	15
14/06/1987	10.2	0	13.1	5	Light	18
15/06/1987	9.2	3.7	7.1	4	Light	25
16/06/1987	10.4	5.6	8.3	6	Light	25
17/06/1987	12.8	0.1	5.3	10	Light	27
18/06/1987	13.0	7.4	3.2	9	Light	24
19/06/1987	14.0	tr	0.4	12	Moderate	33
20/06/1987	12.6	0	7.7	6	Light	17

As previously noted, the **actual** data set has mean temperature for all locations. I changed to maximum temperature for this example for consistency with the textbook.

[Textbook]
(a) Describe the type of data represented by daily total rainfall.

Alison is investigating daily maximum gust. She wants to select a sample of size 5 from the first 20 days in Hurn in June 1987. She uses the first two digits of the date as a sampling frame and generates five random numbers between 1 and 20.
b) State the type of sample selected by Alison.
c) Explain why Alison's process might not generate a sample of size 5.

a ?

b ?

c ?

Hurn

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Date	Daily Max Temp (09-00-0900 C)	Daily Total Rainfall (0900-0900) (mm)	Daily Total Sunshine (0000-2400) (hrs)	Daily Mean Windspeed (0000-2400) (kn)	Daily Mean Windspeed (0000-2400) (Beaufort conversion)	Daily Maximum Gust (0000-2400) (kn)
01/06/1987	15.1	0.6	4.5	7	Light	19
02/06/1987	12.5	4.7	0	7	Light	22
03/06/1987	13.8	tr	5.6	11	Moderate	25
04/06/1987	15.5	5.3	7.8	7	Light	17
05/06/1987	13.1	19	0.5	10	Light	33
06/06/1987	13.8	0	8.9	19	Fresh	46
07/06/1987	13.2	tr	3.8	11	Moderate	27
08/06/1987	12.9	1	1.7	9	Light	19
09/06/1987	11.2	tr	5.4	6	Light	19
10/06/1987	9.2	1.3	9.7	4	Light	n/a
11/06/1987	12.6	0	12.5	6	Light	18
12/06/1987	10.4	0	11.9	5	Light	n/a
13/06/1987	9.6	0	8.6	5	Light	15
14/06/1987	10.2	0	13.1	5	Light	18
15/06/1987	9.2	3.7	7.1	4	Light	25
16/06/1987	10.4	5.6	8.3	6	Light	25
17/06/1987	12.8	0.1	5.3	10	Light	27
18/06/1987	13.0	7.4	3.2	9	Light	24
19/06/1987	14.0	tr	0.4	12	Moderate	33
20/06/1987	12.6	0	7.7	6	Light	17

[Textbook] Calculate:
a) The mean daily maximum temperature for the first five days of June in Hurn in 1987.
b) The median daily total rainfall for the week of 14th June to 20th June inclusive.
c) The median daily total rainfall for the same week in Perth was 19.00mm. Karl states that more southerly countries experience higher rainfall during June. State with a reason whether your answer to part (b) supports this statement.

a ?

b ?

c ?

Questions 3 and 4 in this exercise use the following extracts from the large data set.

LEEMING
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Date	Daily mean temperature (°C)	Daily total rainfall (mm)	Daily total sunshine (hrs)	Daily mean windspeed (kn)
01/06/2015	8.9	10	5.1	15
02/06/2015	10.7	tr	8.9	17
03/06/2015	12.0	0	10.0	8
04/06/2015	11.7	0	12.8	7
05/06/2015	15.0	0	8.9	9
06/06/2015	11.6	tr	5.4	17
07/06/2015	12.6	0	13.9	10
08/06/2015	9.4	0	9.7	7
09/06/2015	9.7	0	12.1	5
10/06/2015	11.0	0	14.6	4

HEATHROW
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Date	Daily mean temperature (°C)	Daily total rainfall (mm)	Daily total sunshine (hrs)	Daily mean windspeed (kn)
01/06/2015	12.1	0.6	4.1	15
02/06/2015	15.4	tr	1.6	18
03/06/2015	15.8	0	9.1	9
04/06/2015	16.1	0.8	14.4	6
05/06/2015	19.6	tr	5.3	9
06/06/2015	14.5	0	12.3	12
07/06/2015	14.0	0	13.1	5
08/06/2015	14.0	tr	6.4	7
09/06/2015	11.4	0	2.5	10
10/06/2015	14.3	0	7.2	10

3 a Work out the mean of the daily total sunshine for the first 10 days of June 2015 in:
- **i** Leeming
- **ii** Heathrow.

b Work out the range of the daily total sunshine for the first 10 days of June 2015 in:
- **i** Leeming
- **ii** Heathrow.

c Supraj says that the further north you are, the fewer the number of hours of sunshine. State, with reasons, whether your answers to parts **a** and **b** support this conclusion.

a?

b?

c?

4 Calculate the mean daily total rainfall in Heathrow for the first 10 days of June 2015. Explain clearly how you dealt with the data for 2/6/2015, 5/6/2015 and 8/6/2015.

?

5 Dominic is interested in seeing how the average monthly temperature changed over the summer months of 2015 in Jacksonville. He decides to take a sample of two days every month and average the temperatures before comparing them.

a Give one reason why taking two days a month might be:
i a good sample size
ii a poor sample size.

b He chooses the first day of each month and the last day of each month. Give a reason why this method of choosing days might not be representative.

c Suggest a better way that he can choose his sample of days.

a?

b?

c?

7 Brian calculates the mean cloud coverage in Leeming in September 1987. He obtains the answer 9.3 oktas. Explain how you know that Brian's answer is incorrect.

?

8 The large data set provides data for 184 consecutive days in 1987. Marie is investigating daily mean windspeeds in Camborne in 1987.

a Describe how Marie could take a systematic sample of 30 days from the data for Camborne in 1987. **(3 marks)**

b Explain why Marie's sample would not necessarily give her 30 data points for her investigation. **(1 mark)**

a?

b?